

Arduino + Neovim + Clangd Setup Notes

Goal

Get proper LSP support for Arduino `.ino` files in Neovim:

- Autocomplete
- Hover (`K`)
- Go to definition (`gd`)
- Diagnostics
- References
- Rename symbol

Initial Attempt: Arduino Language Server

Installed:

- `arduino-cli`
- `clangd`
- `arduino-language-server` (via Mason)

Verified Arduino CLI:

```
arduino-cli version
```

Verified Arduino core:

```
arduino-cli core install arduino:avr
```

Found Arduino CLI config:

```
find ~ -name "arduino-cli.yaml"
```

Result:

```
/Users/vamshidhar/.arduinoIDE/arduino-cli.yaml
```

Tested Arduino Language Server:

```
~/local/share/nvim/mason/packages/arduino-language-server/arduino-language-server
  -cli arduino-cli
  -cli-config /Users/vamshidhar/.arduinoIDE/arduino-cli.yaml
  -clangd clangd
  -fqbn arduino:avr:uno
```

Server started correctly.

Problem Encountered

Inside Neovim:

```
panic: runtime error: invalid memory address
DocumentURI.AsPath()
initializeReqFromIDE()
```

Arduino Language Server crashed during initialization.

Environment:

```
Neovim 0.12.0
arduino-language-server 0.7.7
```

Conclusion:

Arduino Language Server is unstable with this setup.

Decided to switch to pure clangd.

Switching To Clangd

Goal:

Use clangd directly with Arduino compile information.

Generating Compilation Database

Command:

```
arduino-cli compile
  --fqbn arduino:avr:uno
  --only-compilation-database
  --build-path ./build
  .
```

Generated:

```
build/
├─ compile_commands.json
├─ sketch/
└─ ...
```

Important file:

```
build/compile_commands.json
```

This tells clangd:

- include paths
- compiler flags
- board type
- macros

Clangd Configuration

Enabled clangd for:

```
filetypes = {
  "c",
  "cpp",
  "arduino",
}
```

Result:

clangd attached successfully

Arduino Symbols Not Found

Initial errors:

```
Use of undeclared identifier 'digitalWrite'  
Use of undeclared identifier 'OUTPUT'
```

Reason:

Arduino IDE automatically injects:

```
#include <Arduino.h>
```

but clangd does not.

Solution:

Add manually:

```
#include <Arduino.h>  
  
void setup() {  
    pinMode(13, OUTPUT);  
}  
  
void loop() {  
    digitalWrite(13, HIGH);  
}
```

After this:

- Hover works
 - Go to definition works
 - Diagnostics work
-

Treesitter Problem

Neovim 0.12 produced:

```
attempt to call method 'range' (a nil value)
```

Stack trace pointed to:

```
nvim-treesitter
```

Installed Neovim 0.11.7.

Result:

```
Treesitter works correctly
```

Conclusion:

Issue was Neovim 0.12 compatibility.

Final Neovim Version

```
NVIM v0.11.7
```

Final Project Structure

```
test/
├─ build/
│   ├── compile_commands.json
│   ├── test.ino.hex
│   ├── test.ino.elf
│   └─ ...
└─ test.ino
```

Final Arduino Sketch Template

```
#include <Arduino.h>

void setup() {

}

void loop() {

}
```

Final Aliases

Compile:

```
alias ac='arduino-cli compile --fqbn arduino:avr:uno --build-path ./build .'
```

Upload:

```
alias au='arduino-cli upload -p /dev/cu.usbserial-120 --fqbn arduino:avr:uno .'
```

Workflow

Edit code:

```
#include <Arduino.h>
```

Compile:

```
ac
```

Upload:

au

Important Discovery

A separate:

```
arduino-cli compile
  --fqbn arduino:avr:uno
  --only-compilation-database
```

command is NOT needed.

Reason:

Normal compile (`ac`) already generates:

```
build/compile_commands.json
```

Therefore:

```
acc = unnecessary
```

Current Status

Working:

- clangd
- Arduino Uno
- Neovim 0.11.7
- Treesitter
- Hover (`K`)
- Go To Definition (`gd`)
- Diagnostics
- Completion

No Arduino Language Server required.